

Shreya Singh

Software Engineer | Full-Stack AI Developer

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EDUCATION

IIIT Kota

2023 – 2027

B.Tech Computer Science & Engineering | CGPA: 7.81/10

TRAINING PROGRAMS

Trainee

Amazon

Amazon ML Summer School

- Learned ML, Generative AI and LLM fundamentals through sessions led by Amazon scientists.
- Explored how to integrate machine learning into scalable software systems to build intelligent, AI-powered applications.

PROJECTS

LearnSmart – AI Project Analysis and Learning Platform

[GitHub](#) | [Live](#)

MERN Stack, LangChain.js, RAG, OpenRouter API, Mermaid.js, Tree Sitter, GitHub OAuth

- Architected MERN AI project analysis platform supporting 10+ languages using Tree-sitter, AST parsing, and RAG; reduced LLM token usage by 90%+.
- Developed 8+ static analysis modules for architecture, dependencies, DB models, metrics, security, performance, learning resources, and interview prep and project difficulty analysis through XGBoost.
- Implemented 4 AI workflows using LangChain.js, ChromaDB, and NVIDIA/OpenRouter APIs for code explanation, feature analysis, resume extraction, and AI tutoring.
- Generated Mermaid.js architecture diagrams, D3.js knowledge graphs, and GitHub OAuth (Passport.js)

CodeMentor AI – AI Collaborative Coding Platform

[GitHub](#) | [Live](#)

MERN Stack, Socket.IO, OpenRouter API, GitHub OAuth

- Engineered a MERN AI code review and collaborative learning platform supporting 5+ languages, reducing debugging effort by 60% through AI code analysis.
- Leveraged OpenRouter API and Socket.IO for real-time collaborative coding, supporting 10+ users with <100ms latency.
- Delivered collaborative coding, AI review, code analysis, and AI tutoring in a unified platform.
- Orchestrated GitHub OAuth via Passport.js with a scalable Node.js, Express.js, and MongoDB Atlas backend.

AgriVision – AI Crop Health Monitoring System

[GitHub](#) | [Live](#)

MERN Stack, ML (CNN, YOLO, LSTM, Scikit-learn)

- Created a crop disease detection system using YOLO-based computer vision, achieving 95% classification accuracy.
- Trained 3 ML models (CNN, YOLO, LSTM) for crop disease classification, weather prediction, and soil analysis.
- Engineered CNN-based soil health module using soil indices and weather data for crop fit recommendation.
- Designed real-time prediction pipeline with <2s response via 5+ scalable REST APIs; React UI improving usability by 30%.

TECHNICAL SKILLS

Languages: C++, JavaScript, SQL, Python(Numpy, Pandas, Matplotlib, Seaborn), C

Frontend: React.js, Socket.IO Client, Axios, Material UI, HTML, CSS, Tailwind CSS, Responsive Design

Backend: Node.js, Express.js, WebSockets, OAuth, REST APIs, JWT Authentication, API Integration, API Design

Databases: MongoDB, MySQL

AI/ML: AI/LLM Integration, Generative AI, RAG, LangChain.js, ML Fundamentals (Python, Scikit-learn)

Development Tools & Cloud: Git, GitHub, Postman, Supabase, VS Code, GitHub Actions

CS Fundamentals: Data Structures & Algorithms, System Design, OOP, DBMS, OS, Computer Networks

ACHIEVEMENTS & LEADERSHIP

Smart India Hackathon 2025 – Grand Finalist | Finalist – NextGen AI'26

Selected for the 36-hour onsite hackathon among 86,000+ participants, solving the MathWorks problem statement.

Amazon ML Summer School 2026

Selected among the top 2.2% of 134,000+ applicants nationwide

Flipkart Grid 8.0 Semi-Finalist

Selected from over 165,000 participants nationwide for semi final round.

CodeChef 3 Star (1600+ rating) | 500+ DSA problems solved across various platforms

CodeChef Profile | LeetCode Profile

Certifications: IBM SkillBuild Data Fundamentals (IBM) | Professional Communication (IIIT Kota)